VLSI Devices
Lecture 14

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The second half

- We have 11 lectures (L14~L24) left.
- Final exam: **June 10**
  - Recorded video for L19

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	14		15		1	2
3	4 16	5	6 17	7	8	9
10	11 17	12	13 18	14	15	16
17	18 19	19	20 20	21	22	23
24	25	26	27 21	28	29	30
31						

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 22	2	3 23	4	5	6
7	8 24	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

L14

Let's introduce a factor, m.

- It is given as

$$m = 1 + \frac{\sqrt{\epsilon_{si} q N_a / (4\phi_B)}}{C_{ox}}$$ Taur, Eq. (5.28)

  - From the maximum depletion width,

$$W_{dm} = \sqrt{\frac{4\epsilon_{si}\phi_B}{qN_a}}$$ Taur, Eq. (4.24)

  - An alternative form is available,

$$m = 1 + \frac{C_{dm}}{C_{ox}} = 1 + \frac{3t_{ox}}{W_{dm}}$$ Taur, Eq. (5.29)

Its physical meaning

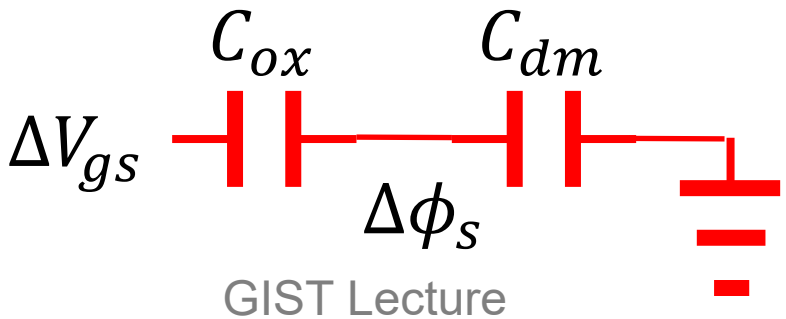
- Serial capacitors give $\frac{C_{ox}C_{dm}}{C_{ox}+C_{dm}}$.
  - Charge across the oxide capacitor:

$$\frac{C_{ox}C_{dm}}{C_{ox}+C_{dm}}\Delta V_{gs} = C_{dm}\Delta\phi_s$$

  - Therefore,

$$m = \frac{C_{ox}+C_{dm}}{C_{ox}} = \frac{\Delta V_{gs}}{\Delta\phi_s}$$

  - m should be kept close to one.



Saturation current

- Maximum value of I_d at a given V_{gs}
  - Recall that

$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs}-V_t\right)V_{ds} - \frac{m}{2}C_{ox}V_{ds}^2\right\}$$ Taur, Eq. (5.27)

  - When $V_{ds} = V_{dsat} = \frac{V_{gs}-V_t}{m}$,

$$I_d = I_{dsat} = \mu_{eff}C_{ox}\frac{W}{L}\frac{\left(V_{gs}-V_t\right)^2}{2m}$$ Taur, Eq. (5.31)

Pinch-off

- For $\phi_s = 2\phi_B + V$, the Tylor expansion gives

$$Q_i = -C_{ox}\left(V_{gs} - V_{fb} - 2\phi_B - V\right) + \sqrt{4\epsilon_{si} q N_a \phi_B} + \sqrt{\frac{\epsilon_{si} q N_a}{4\phi_B}} V$$

$$Q_i = -C_{ox}\left(V_{gs} - V_t - V - \frac{1}{C_{ox}}\sqrt{\frac{\epsilon_{si} q N_a}{4\phi_B}} V\right) = C_{ox}\left(V_{gs} - V_t - mV\right)$$

Taur, Eq. (5.32)

  - At $V_{ds} = \frac{V_{gs} - V_t}{m}$ (on-set of saturation), the surface channel vanishes at the drain end of the channel.

Subthreshold current (1)

- Subthreshold region where $V_{gs} < V_t$
  - Recall that

$$-Q_s = \sqrt{2\epsilon_{si} k_B T N_a}\left[\frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2}\exp\left(\frac{q}{k_B T}(\phi_s - V)\right)\right]^{1/2}$$

Taur, Eq. (5.34)

Small in the subthreshold region

  - Its Taylor expansion

$$-Q_s \approx \sqrt{2\epsilon_{si} k_B T N_a}\left[\sqrt{\frac{q\phi_s}{k_B T}} + \frac{1}{2}\sqrt{\frac{k_B T}{q\phi_s}}\frac{n_i^2}{N_a^2}\exp\left(\frac{q}{k_B T}(\phi_s - V)\right)\right]$$

Subthreshold current (2)

- The second term for the inversion charge

$$-Q_i \approx \sqrt{\frac{\epsilon_{si} q N_a}{2\phi_s}} \frac{k_B T}{q} \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T}(\phi_s - V)\right)$$

Taur, Eq. (5.35)

  - In this case, ϕ_s is a function of V_{gs} only. Recall that

$$I_d = \mu_{eff} \frac{W}{L} \int_0^{V_{ds}} [-Q_i(V)] dV$$

Taur, Eq. (5.10)

  - Then, we have

$$I_d = \mu_{eff} \frac{W}{L} \sqrt{\frac{\epsilon_{si} q N_a}{2\phi_s}} \left(\frac{k_B T}{q}\right)^2 \left(\frac{n_i}{N_a}\right)^2 \exp\left(\frac{q\phi_s}{k_B T}\right) \left(1 - \exp\left(-\frac{qV_{ds}}{k_B T}\right)\right)$$

Taur, Eq. (5.36)

Subthreshold current (3)

- Since $m = \frac{\Delta V_{gs}}{\Delta \phi_s}$, $V_{gs} - V_t = m(\phi_s - 2\phi_B)$.
  - Then,

$$\exp\left(\frac{q\phi_s}{k_B T}\right) = \exp\left(\frac{q(V_{gs} - V_t)}{m k_B T}\right) \exp\left(2\frac{q\phi_B}{k_B T}\right)$$

  - From the above expression,

$$I_d = \mu_{eff} \frac{W}{L} \sqrt{\frac{\epsilon_{si} q N_a}{4\phi_B}} \left(\frac{k_B T}{q}\right)^2 \exp\left(\frac{q(V_{gs} - V_t)}{m k_B T}\right) \left(1 - \exp\left(-\frac{q V_{ds}}{k_B T}\right)\right)$$

Taur, Eq. (5.38)

Subthreshold slope (1)

- I_d is independent of V_{ds}, when $V_{ds} \gg k_B T/q$.

$$I_d = \mu_{eff} \frac{W}{L} \sqrt{\frac{\epsilon_{si} q N_a}{2\phi_s}} \left(\frac{k_B T}{q}\right)^2 \exp\left(\frac{q(V_{gs} - V_t)}{m k_B T}\right)$$

  - Its gate voltage dependence is very important.

$$\log_{10} I_d = (a\ constant) + \frac{q(V_{gs} - V_t)}{m k_B T} \log_{10} e$$

$$\frac{d(\log_{10} I_d)}{dV_{gs}} = \frac{q}{m k_B T} \log_{10} e$$

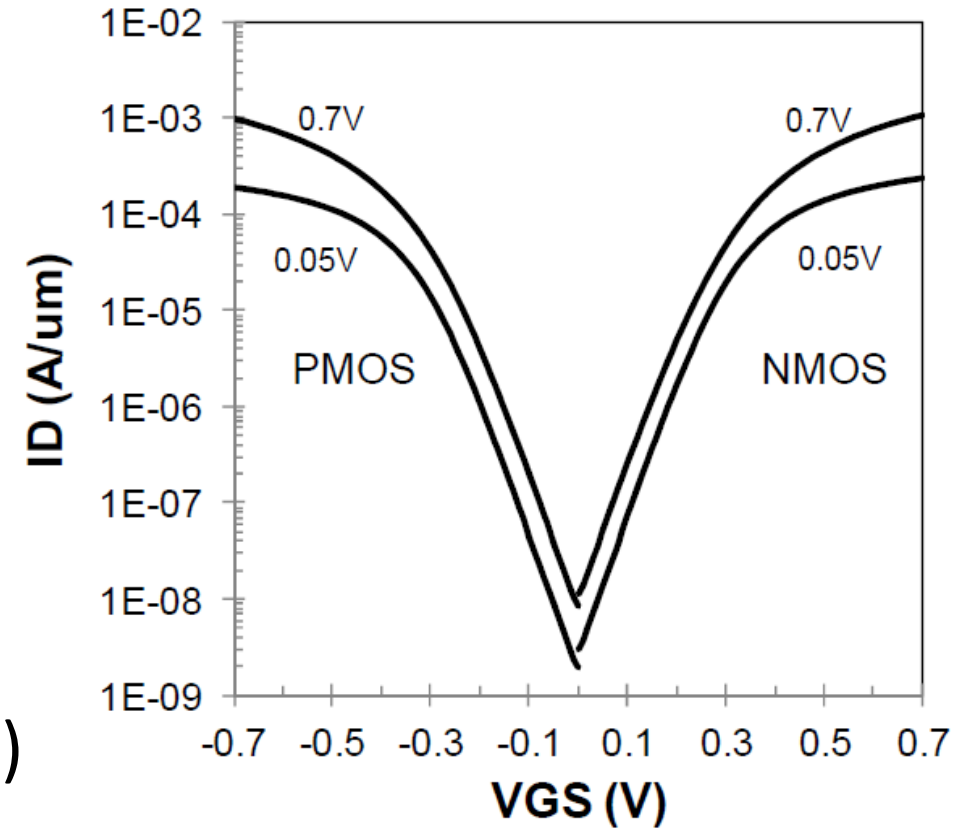
Subthreshold slope

$$S = \left(\frac{d(\log_{10} I_d)}{dV_{gs}}\right)^{-1} = \frac{m k_B T}{q} \ln 10$$

Taur, Eq. (5.40)

Subthreshold slope (2)

- At 300 K, $\frac{k_B T}{q} \ln 10$ is 60 mV/dec.
  - Note that m is larger than 1.



Subthreshold behavior
(Natarajan, IEDM 2024)

Substrate bias (1)

- Assume that the substrate is biased with V_{bs}. (For NMOSFETs, $V_{bs} < 0$)
  - Recall that ($\frac{dV}{d\phi_s} \approx 1$)

$$I_d \approx \mu_{eff} \frac{W}{L} \left\{ C_{ox}\left(V_{gs} - V_{fb}\right)\phi_s - \frac{1}{2} C_{ox}\phi_s^2 - \frac{2}{3}\sqrt{2\epsilon_{si} q N_a}\phi_s^{1.5} \right\}\Bigg|_{\phi_{s,s}}^{\phi_{s,d}}$$

  - Use $V_{gs} \Rightarrow V_{gs} - V_{bs}$, $\phi_{s,s} \Rightarrow 2\phi_B - V_{bs}$, and $\phi_{s,d} \Rightarrow 2\phi_B - V_{bs} + V_{ds}$.

Substrate bias (2)

- Then, we have

$$I_d = \mu_{eff}\frac{W}{L}\left[C_{ox}\left(V_{gs} - V_{fb} - 2\phi_B - \frac{1}{2}V_{ds}\right)V_{ds} - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}(2\phi_B - V_{bs} + V_{ds})^{1.5} + \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}(2\phi_B - V_{bs})^{1.5}\right]$$

Taur, Eq. (5.58)

  - First-order expansion yields

$$I_d = \mu_{eff}\frac{W}{L}\left[C_{ox}\left(V_{gs} - V_{fb} - 2\phi_B\right) - \sqrt{2\epsilon_{si}qN_a}(2\phi_B - V_{bs})^{0.5}\right]V_{ds}$$

Substrate bias (3)

- Therefore, at low drain voltages,

$$I_d = \mu_{eff} C_{ox} \frac{W}{L} \left[\left(V_{gs} - V_{fb} - 2\phi_B \right) - \frac{1}{C_{ox}} \sqrt{2\epsilon_{si} q N_a (2\phi_B - V_{bs})} \right] V_{ds}$$

  - It means that

$$V_t = V_{fb} + 2\phi_B + \frac{\sqrt{2\epsilon_{si} q N_a (2\phi_B - V_{bs})}}{C_{ox}}$$

Taur, Eq. (5.59)

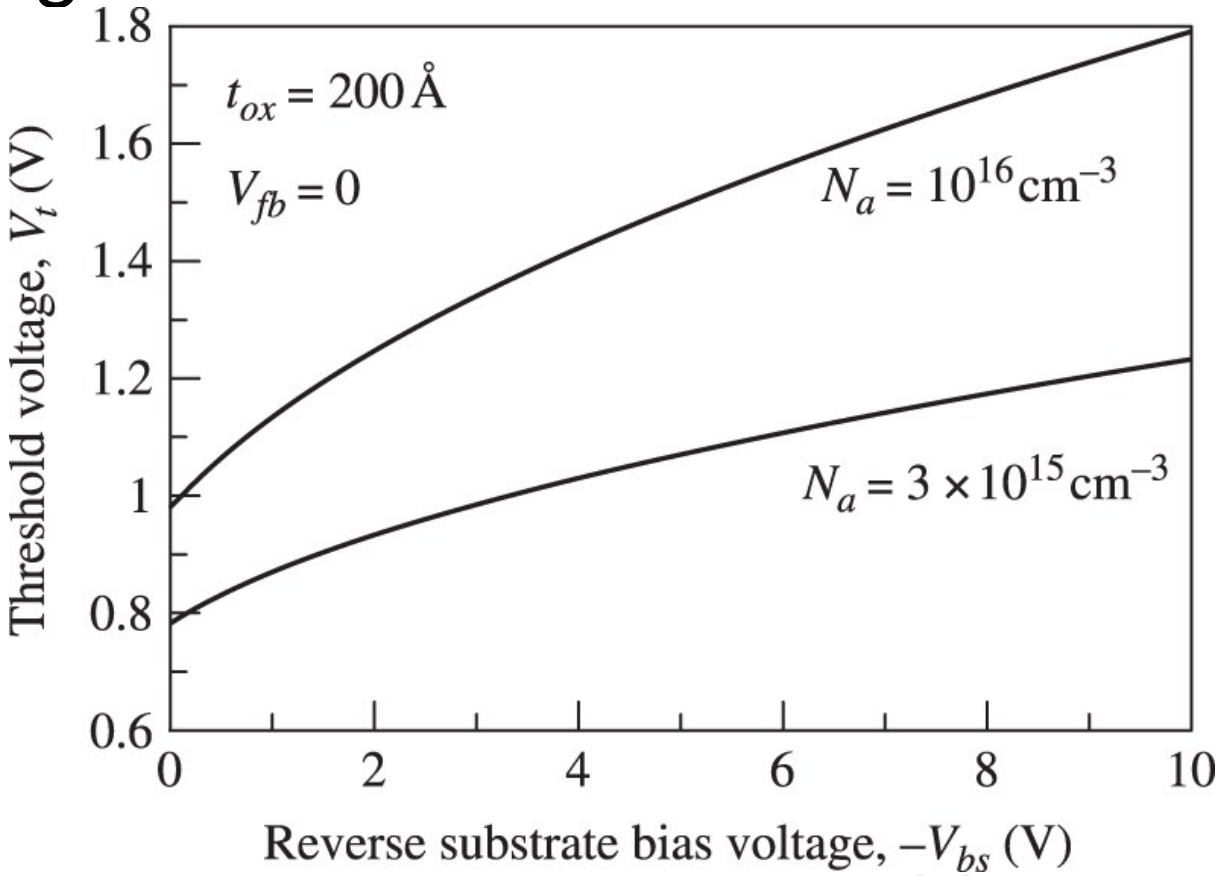
  - Substrate sensitivity

$$\frac{dV_t}{d(-V_{bs})} = \frac{\sqrt{\epsilon_{si} q N_a}}{C_{ox}\sqrt{2(2\phi_B - V_{bs})}}$$

Taur, Eq. (5.60)

Substrate bias (4)

- A reverse substrate bias is to widen the bulk depletion region and raise the threshold voltage.



Threshold voltage variation with reverse substrate bias for two uniform substrate doping concentration (Taur, Fig. 5.20)

Thank you!